

One Pager



HoneyShield — Active Cyber Deception & Honeypot System

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Main Points

Silent reroute • 4 trap types: SQLi / Brute-Force / XSS / Honeypot • Data Bomb / HTTP Trait / Infinite Redirects • Live attack map • Attacker fingerprinting (IP, OS, risk score) • Air-gapped DB

Solution

An active honeypot that silently redirects suspicious traffic to a convincing fake environment. Attackers are trapped (SQLi tarpit, brute-force fake login, XSS sandbox, honeypot files), exhausted (100 GB data bomb, HTTP Trait, infinite redirects), profiled, and reported live to a secured Blue Team admin dashboard.

Problem

Traditional security tools block known threats but are blind to slow, methodical attackers probing networks carefully. There is no standard mechanism to detect unknown intruders early, waste their time and resources, and gather intelligence on their tactics — without alerting them that they have been discovered.

The Methodology

Nginx terminates SSL and injects X-Forwarded-For headers. The Express middleware performs O(1) IP-reputation lookup and regex scanning (SQLi + XSS). Detected threats are silently rerouted via req.url mutation — no HTTP 302, under 50 ms, invisible to the attacker. Real users go through MVC/EJS with bcrypt/MongoDB. Attacker data is stored in a fully isolated Mongoose connection pool.

System Domain

Cybersecurity — Active Network Deception, Honeypot Technology & Real-Time Threat Intelligence

Modules

Gatekeeper Middleware — IP blacklist (O(1)) + regex SQLi / XSS detection
Safe Zone — Real app: MVC / EJS / bcrypt / MongoDB Atlas
Deception Engine — SQLi tarpit, brute-force fake login, XSS sandbox, honeypot files
Telemetry & Logging — Winston + MongoDB; wasted-time metrics; risk scoring
Admin Dashboard — React SPA: live logs, stats panel, Leaflet.js attack map

Interfaces

Nginx Reverse Proxy (HTTP/HTTPS — ports 80/443, SSL termination)
Express REST API (admin CRUD, IP ban, 2FA-protected endpoints)
Socket.io WebSocket (real-time event push to the dashboard)
React SPA (Admin UI with Leaflet.js geographic attack map)
MongoDB Atlas (two isolated pools: real data / attacker data)
NetIQ / OAuth 2FA (JWT stored in HttpOnly cookie)

Description of System Components

Node.js / Express behind an Nginx reverse proxy. A central middleware pipeline classifies every request: clean users go to the Safe Zone (real MVC app — EJS, bcrypt, MongoDB); detected attackers are silently rerouted to the Deception Engine (honeypot traps + fakerLogin.js dynamic fake data). All attacker events are stored in an isolated MongoDB instance and broadcast live via Socket.io to the React admin dashboard.

Defining the Goals of the Project

1. Detect & classify malicious traffic in real time. 2. Silently reroute attackers to a fake environment without alerting them. 3. Exhaust attacker resources (CPU, bandwidth, time). 4. Profile attackers: IP, device fingerprint, attack pattern & riskscore. 5. Provide the Blue Team with a live secured admin dashboard.

Summary and Conclusions

HoneyShield is an active cyber-deception platform. It lures attackers into a realistic fake environment, traps and exhausts their resources, profiles their identity and tactics, and streams live threat intelligence to the Blue Team — while the real application remains fully protected and completely invisible to the intruder.

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maximum one page, the font must be used: Calibri